Chapter 3, Part II: Prices and Foreclosures

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Federal Reserve Bank of Boston, Harvard and NBER

2410, April 6, 2017

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Prepared for ec2410 at Harvard, Spring 2017.
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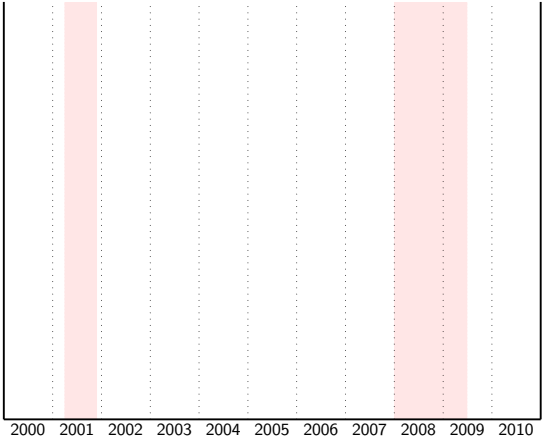
Course Overview



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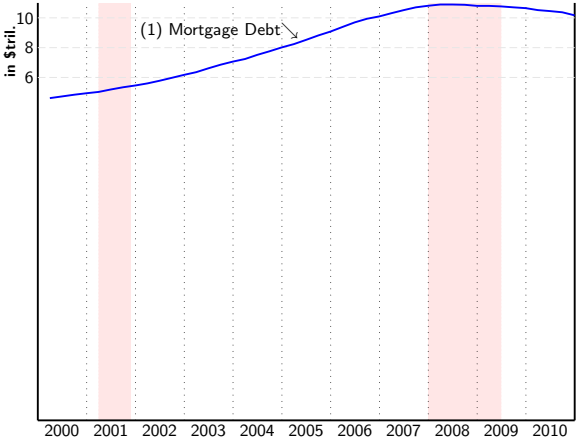

- (1) Debt Growth

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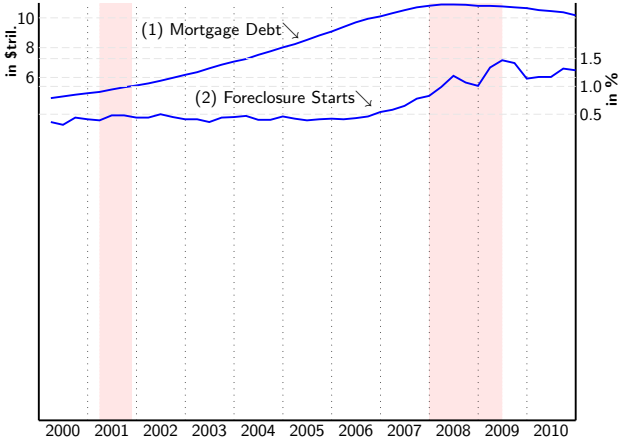

- (1) Debt Growth
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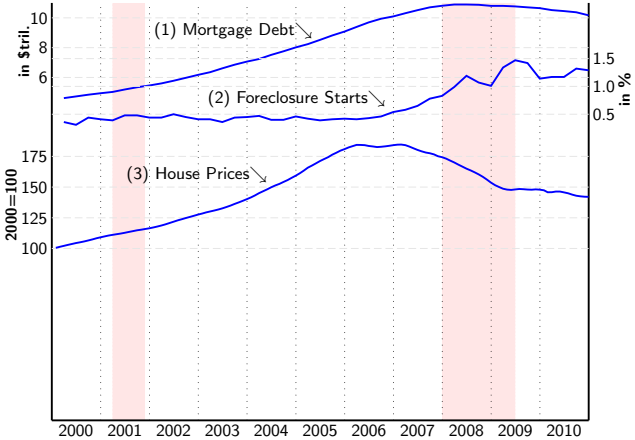

- (1) Debt Growth
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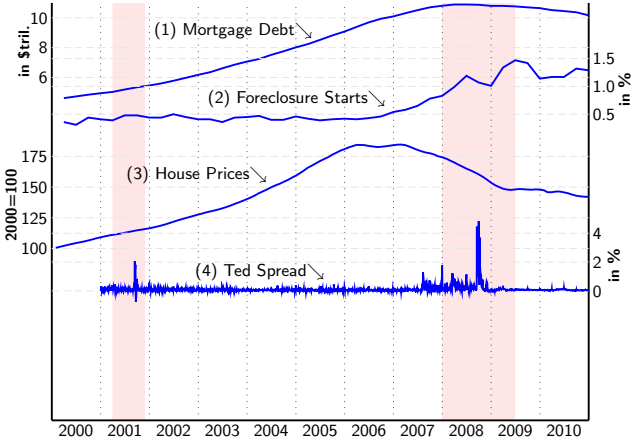

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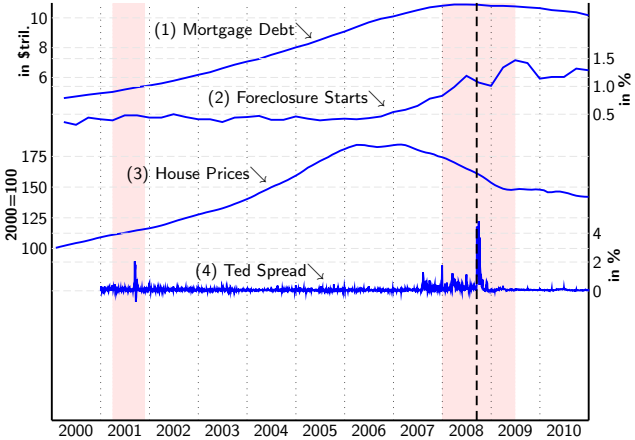

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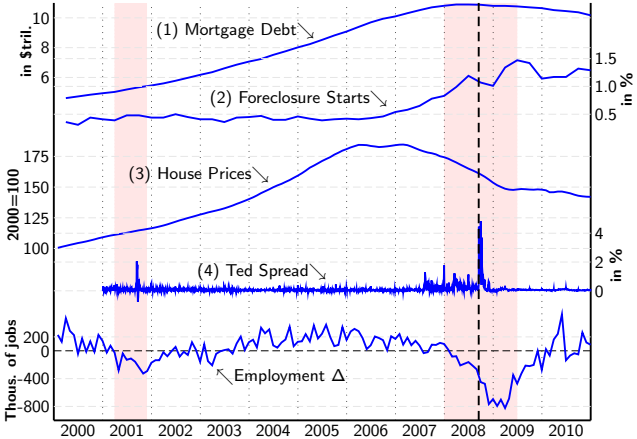

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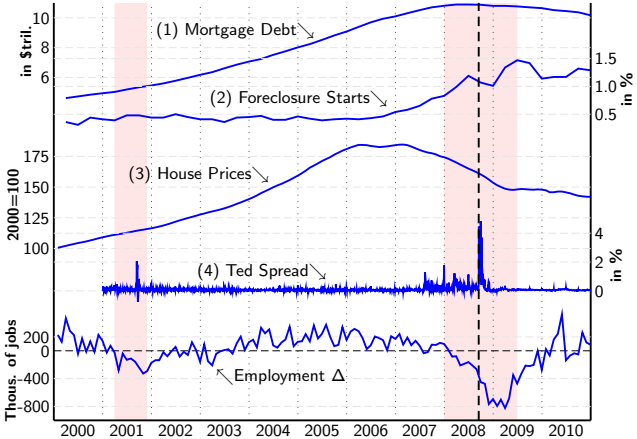

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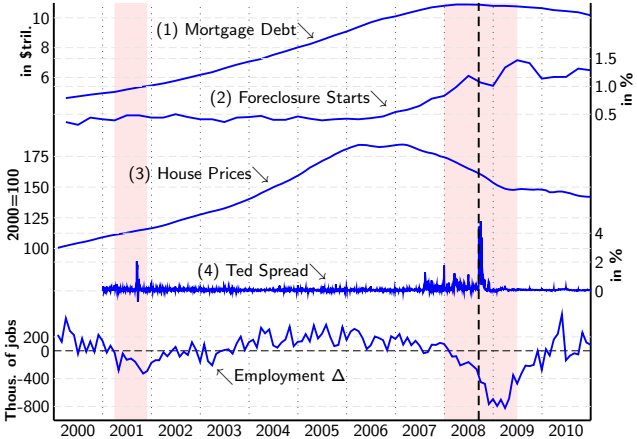

- (1) Debt Growth
- (2) Default
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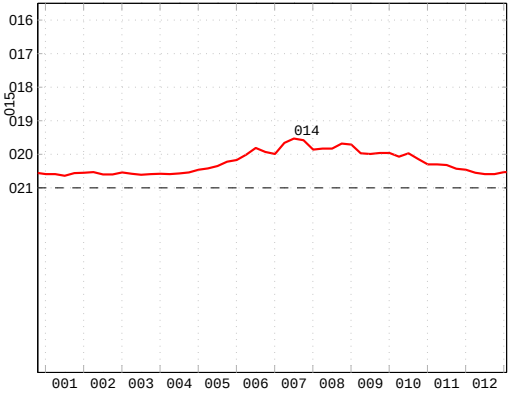
- (5) Institutions
- (6) Expectations

Foreclosures and House Prices

- Theory
  - Prices ⇒ Foreclosures
  - Foreclosures ⇒ Prices?
- Two stories
  1. Supply effect: "Flood of properties"
  2. Externalities
- Supply effect?
  - Homeowner moves to new home
  - Lender sells house
- Does it affect supply or demand?

$$\underbrace{h = \frac{\beta}{1+\beta}\frac{y_0}{P_0 - \frac{P_1}{R_0}}}_{\text{Demand}} = \underbrace{\overline{h}}_{\text{Supply}}$$

- Affects properties "on the market"



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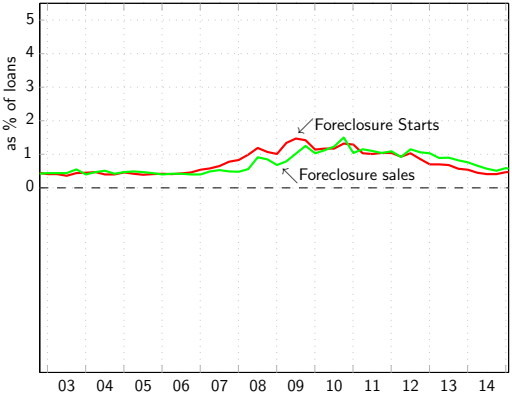

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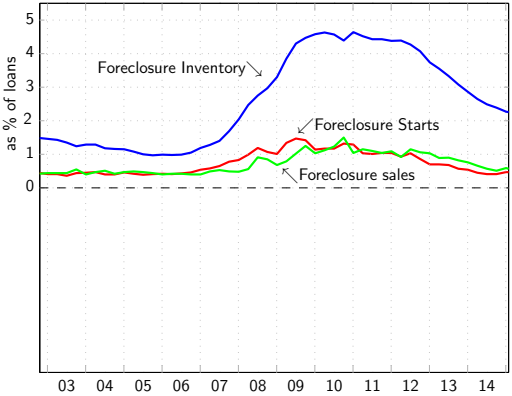

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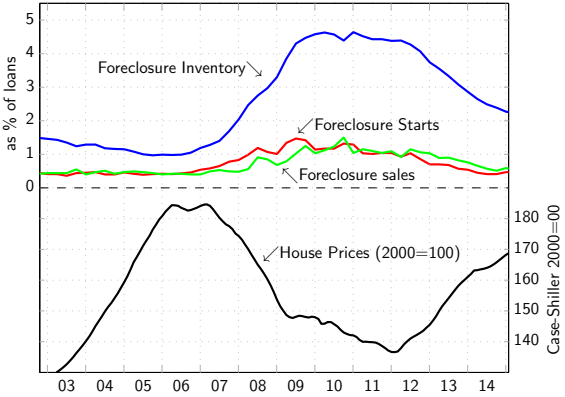

Guren and McQuade (2013)

- More "Realistic" Model:
  - Search model
  - Foreclosure ⇒ Homeowner leaves housing market – No rental market (later version changed)
- Two effects:
  1. Choosey Buyer: Foreclosure seller is outside option in search model – almost no effect
  2. Market Tightness: Fewer buyers – big effect
- Back to simple model:

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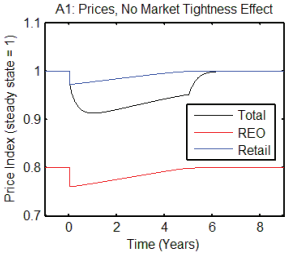
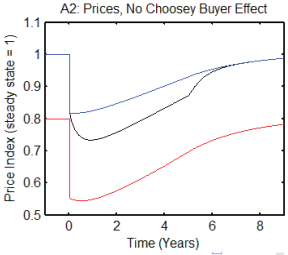
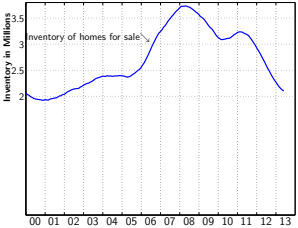
- Foreclosures reduce *demand*.





Evidence on "Supply Effect"

- Major identification problem
- Anenberg and Kung (2014)
  - Show that foreclosure listings lead nearby sellers to lower prices
  - Also true for non-foreclosure listings
  - Overall, listings collapsed in the bust –



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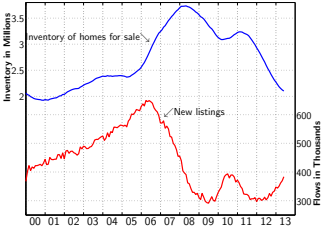

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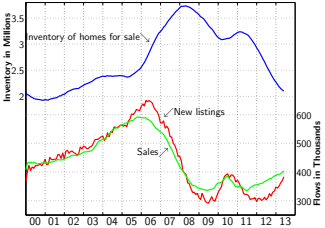

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  - Overall, listings collapsed in the bust –
- Mian, Sufi, and Trebbi (2015)
  - Exploited state level differences
  - Not randomly assigned
  - Regression discontinuity?
  - Doesn't work

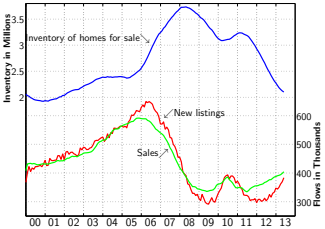
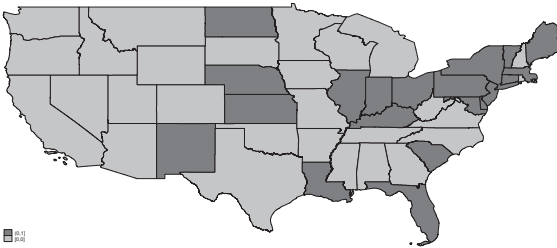
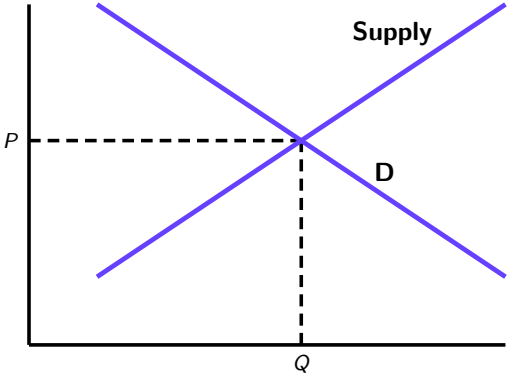


Figure 2
States with Judicial Foreclosure Requirement
States shaded in dark gray require judicial foreclosure. The data come from RealtyTrac.com and are available at: http://www.realtytrac.com/foreclosure-laws/foreclosure-laws-comparison.asp



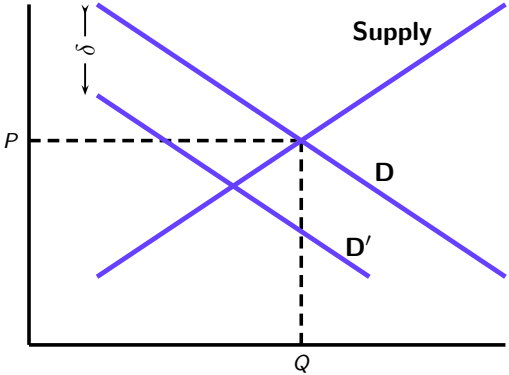
A demand shock

- Equilibrium in a housing market.



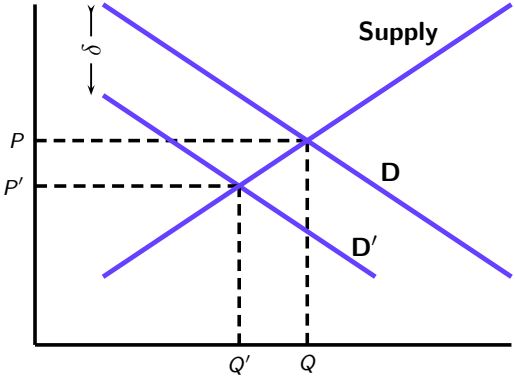
A demand shock

- Equilibrium in a housing market.
- A negative demand shock



A demand shock

- Equilibrium in a housing market.
- A negative demand shock
- Lowers prices and quantities



A demand shock

- Equilibrium in a housing market.
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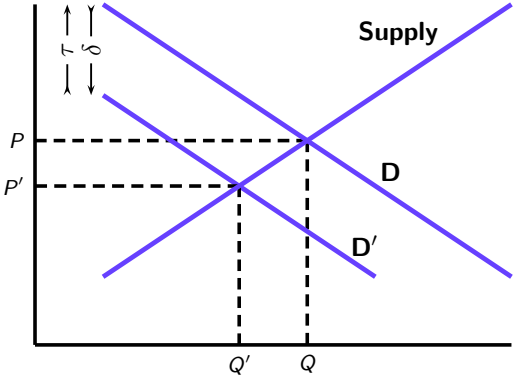
- Lowers prices and quantities
- Suppose we introduce a subsidy that shifts demand back to D'



A demand shock

- Equilibrium in a housing market.
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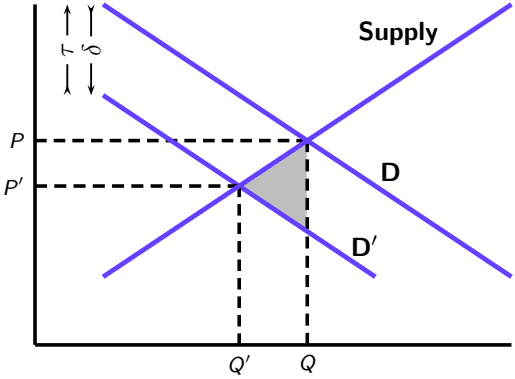
- Standard theory says we will create a deadweight loss



A demand shock

- Equilibrium in a housing market.
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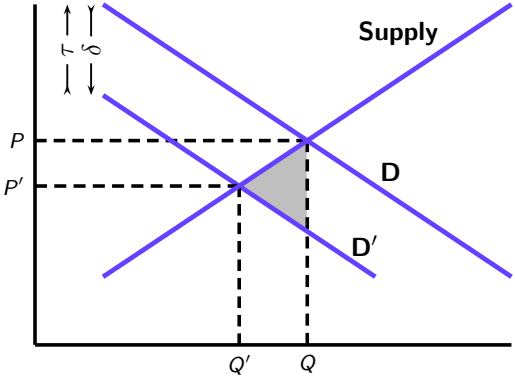
- Suppose we introduce a subsidy that shifts demand back to D'
- Standard theory says we will create a deadweight loss
- Buyers value the house less than sellers



A Supply Shock

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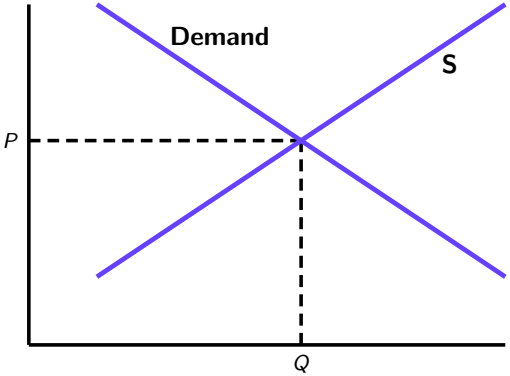
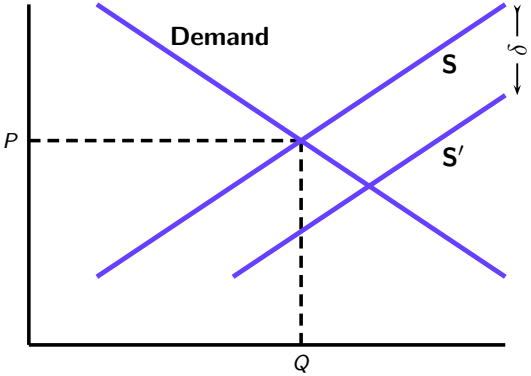


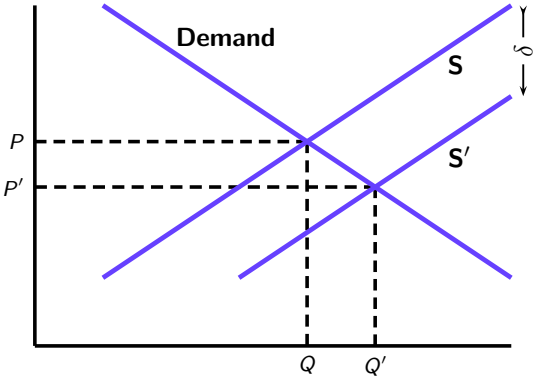
A Supply Shock

- Equilibrium in a housing market.
- A positive supply shock



A Supply Shock

- Equilibrium in a housing market.
- A positive supply shock
- Lowers prices and increases quantity



A Supply Shock

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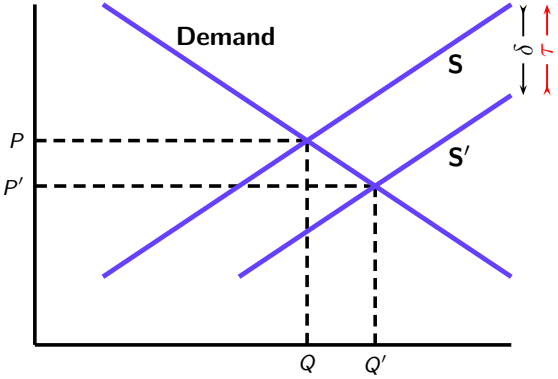
- Suppose we introduce a tax that shifts demand back to S



A Supply Shock

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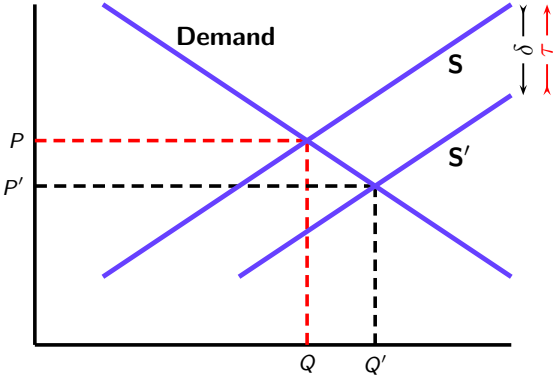
- Suppose we introduce a tax that shifts demand back to S
- Prices go up!



A Supply Shock

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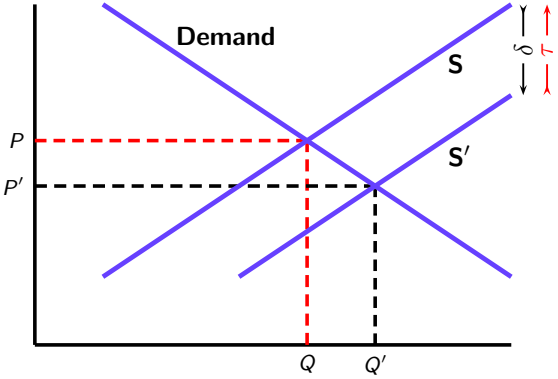
- Prices go up!
- Are higher prices good news?



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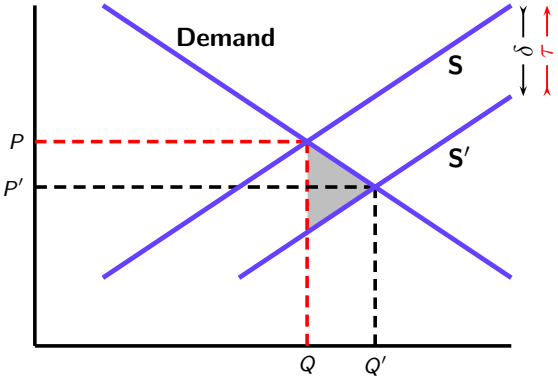

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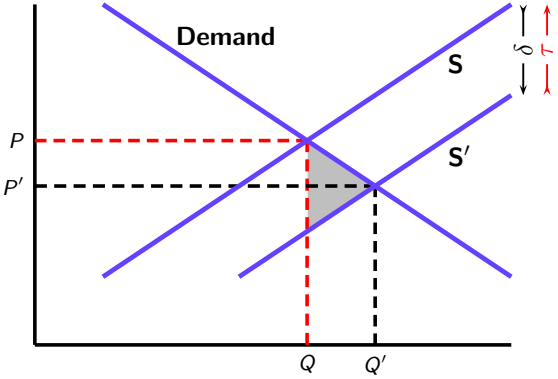
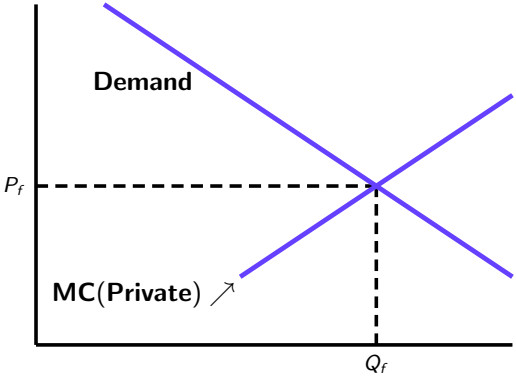


Externalities: Market for Foreclosures

- Equilibrium in the market for foreclosed property.



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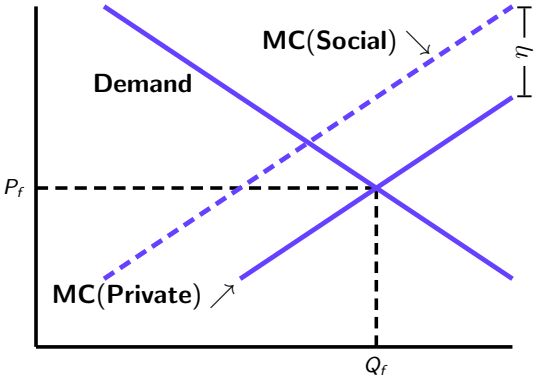

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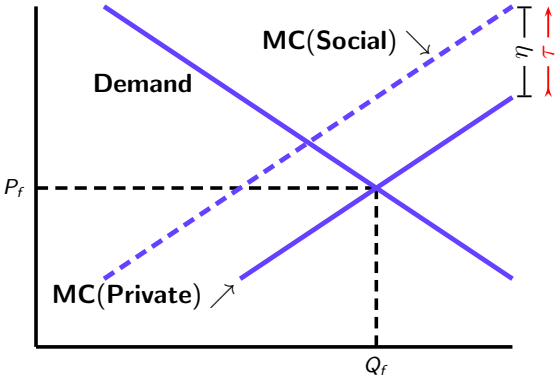

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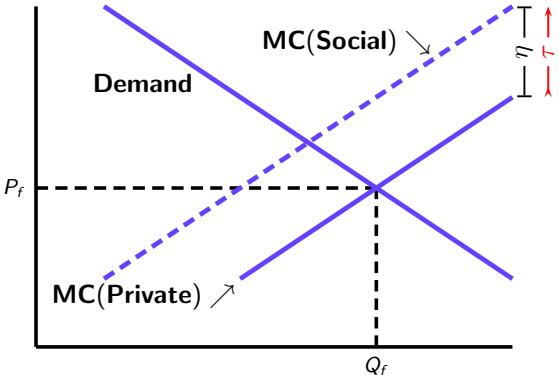

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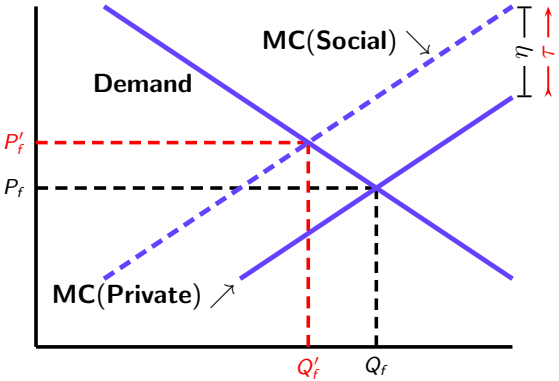

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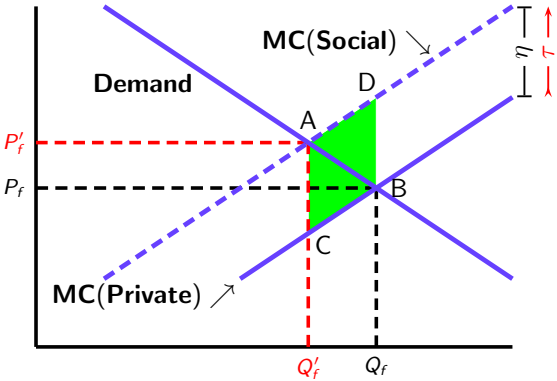
- Social gain ADBC $((Q_f - Q) \times \eta)$



Externalities: Market for Foreclosures

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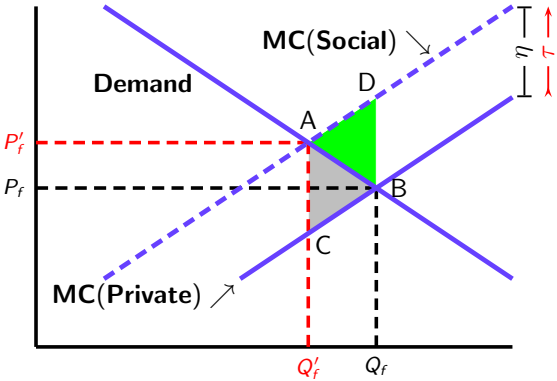

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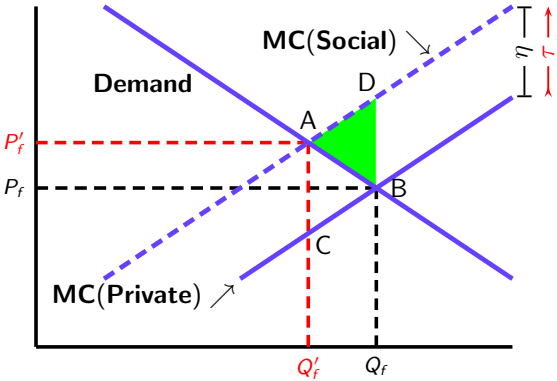

Externalities: Market for Nearby Properties

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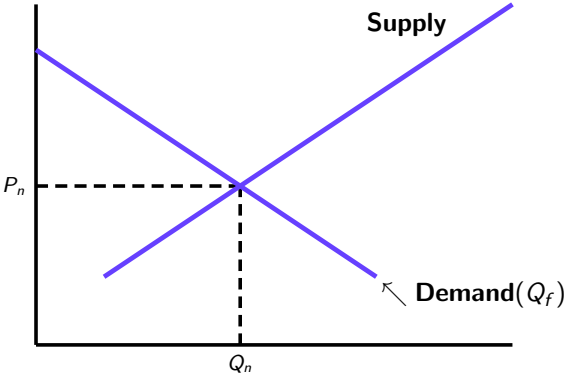

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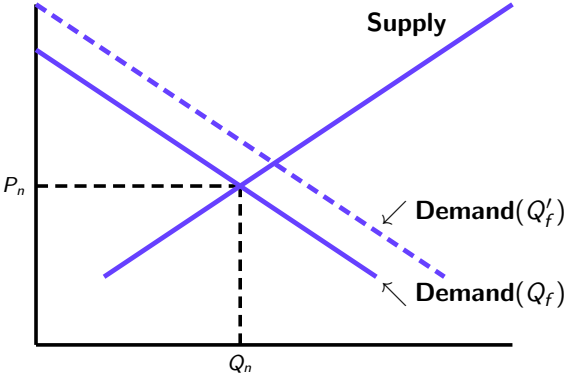

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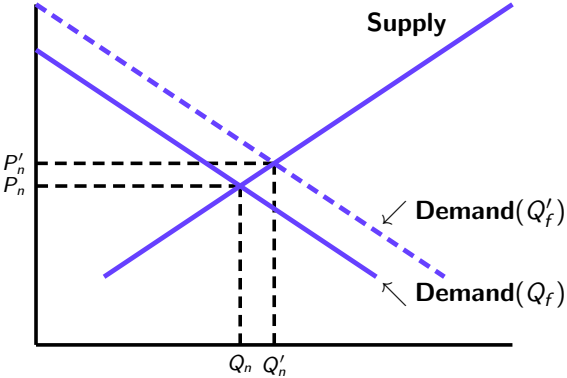

Externalities: Market for Nearby Properties

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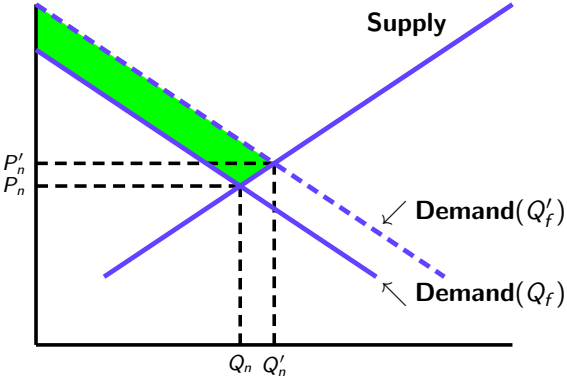
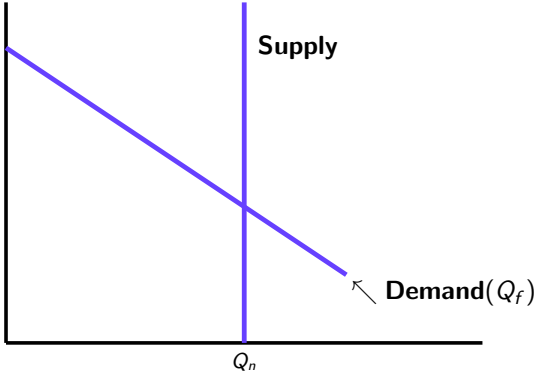


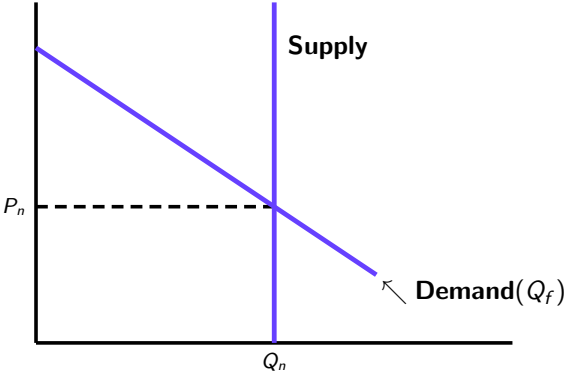
Externalities: Market for Nearby Properties

- Supply of property is total number of houses



Externalities: Market for Nearby Properties

- Supply of property is total number of houses
- Determines the price



- From a Federal Reserve White Paper

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> *In particular, foreclosures can be a costly and inefficient way to resolve the inability of households to meet their mortgage payment obligations because they can result in "deadweight losses," or costs that do not benefit anyone, including the neglect and deterioration of properties that often sit vacant for months (or even years) and the associated negative effects on neighborhoods.*[2] *These deadweight losses compound the losses that households and creditors already bear and can result in further downward pressure on house prices.*

[2] See, for example, John Y. Campbell, Stefano S. Giglio, and Parag P. Pathak (2011), "Forced Sales and House Prices," *American Economic Review*, vol. 101 (August), pp. 2108-31; and Dan Immergluck and Geoff G. Smith (2006), "The External Costs of Foreclosure: The Impact of Single-Family Mortgage Foreclosures on Property Values," *Housing Policy Debate*, vol. 17 (1), pp. 57-80.

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- Let me illustrate with an example.

Identification problem

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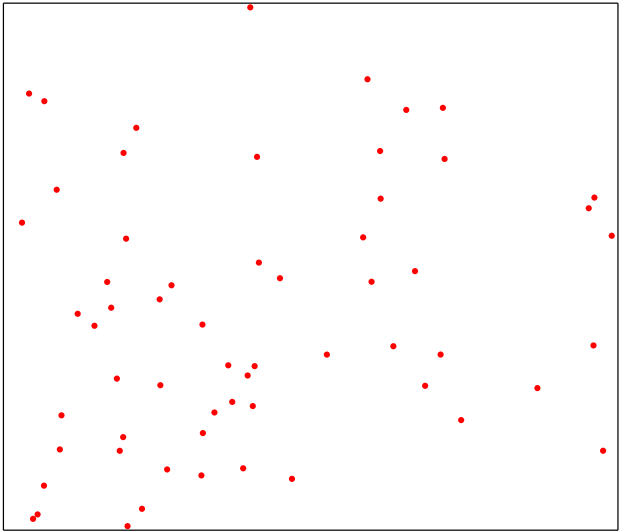

- We have a bunch of foreclosures

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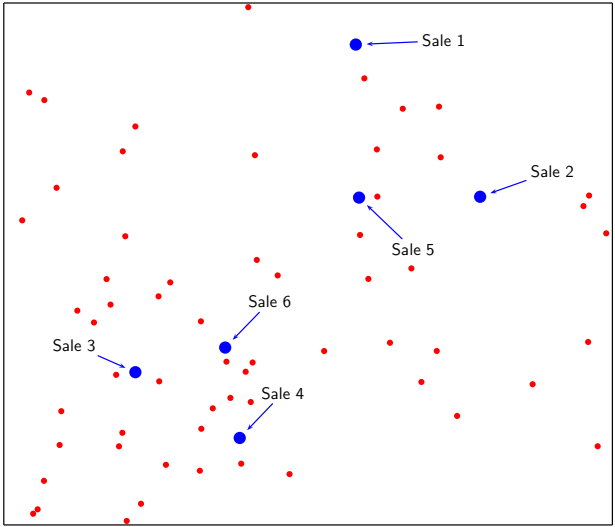

- We have a bunch of foreclosures
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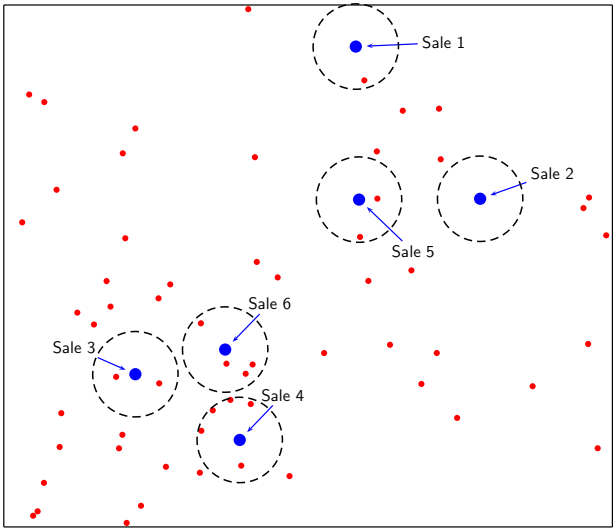

- We have a bunch of foreclosures
- And a bunch of sales
- Look in a radius around the sales and count foreclosures

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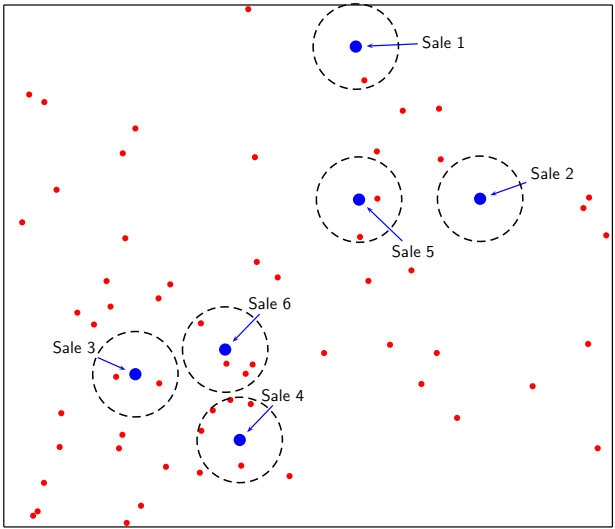

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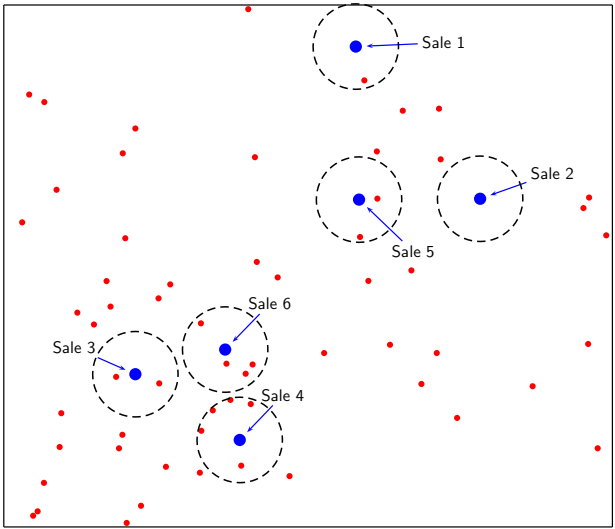

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Sale	# F's	P
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2	0	5
3	2	3
4	5	3
5	2	5
6	4	3

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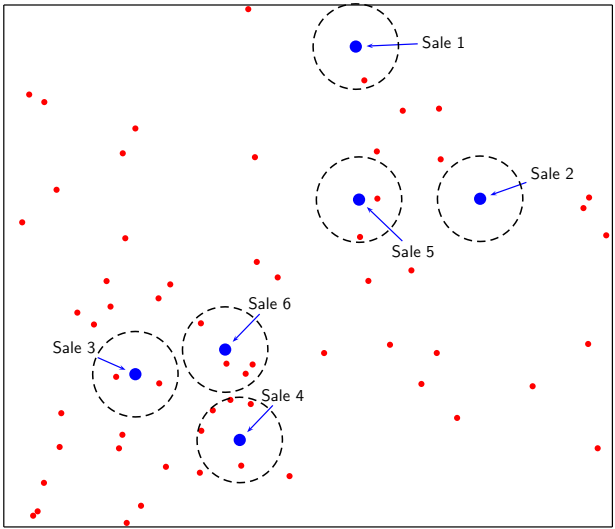

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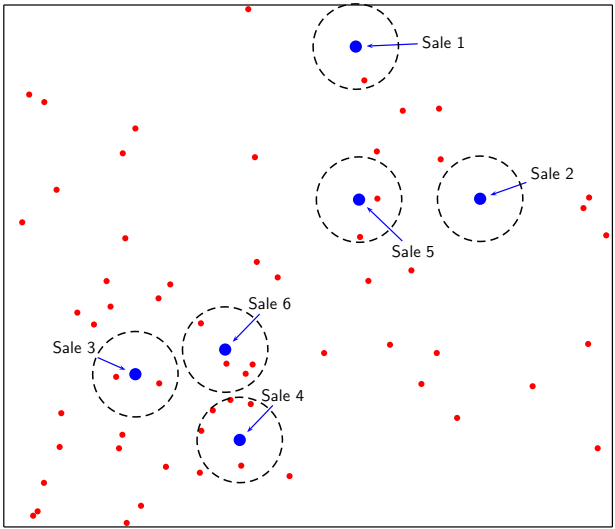

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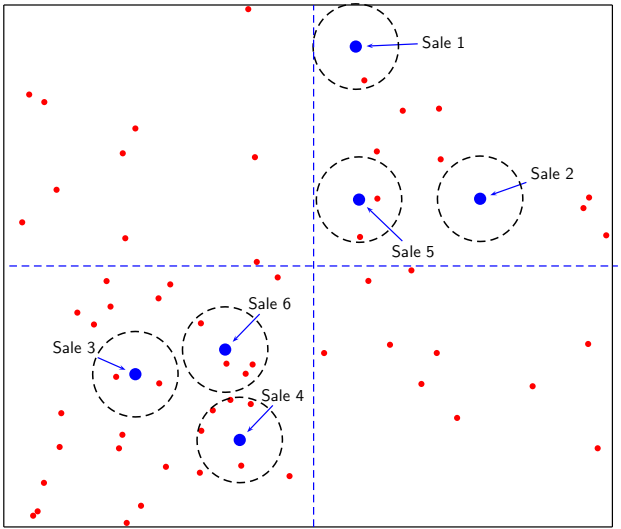

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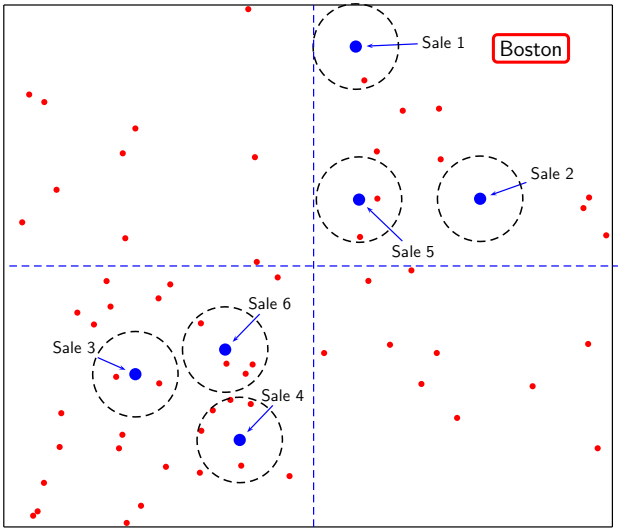

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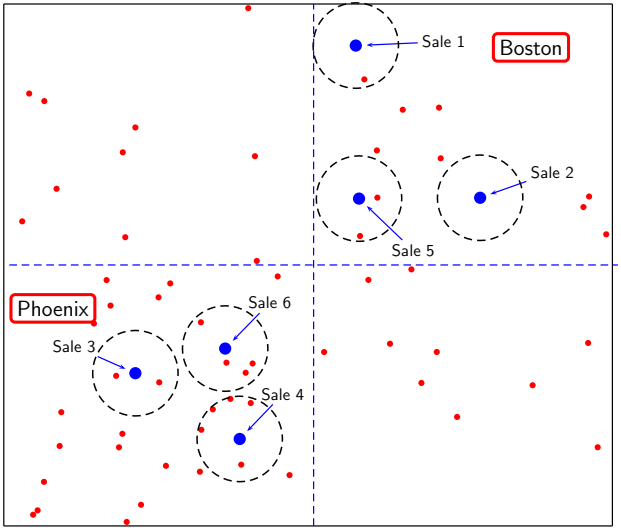

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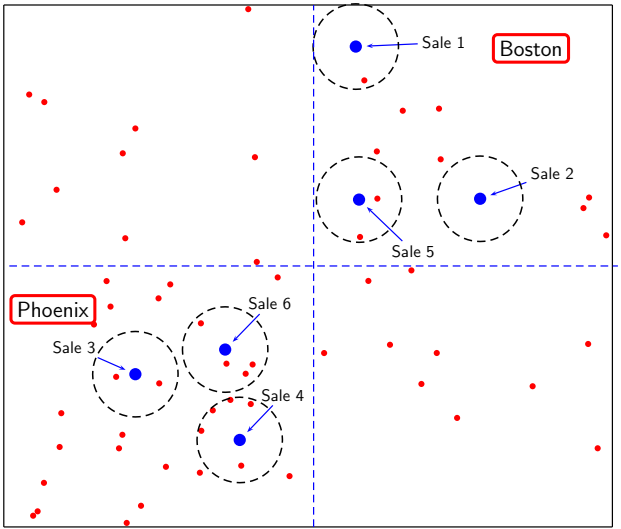

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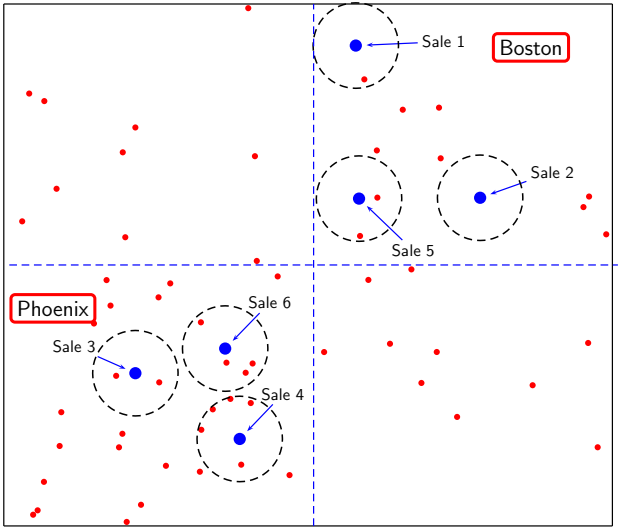

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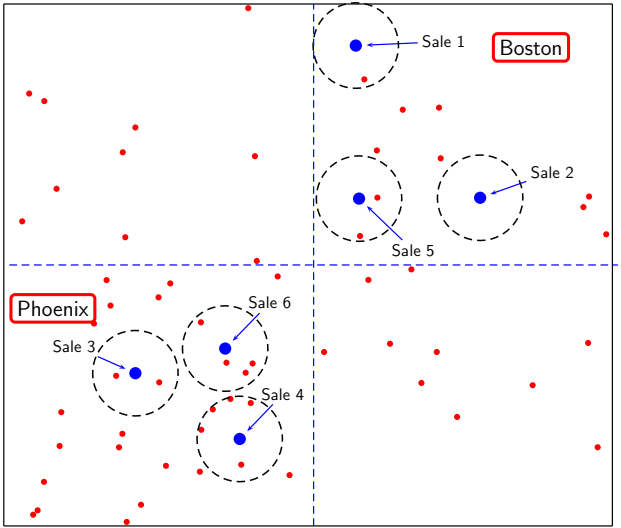

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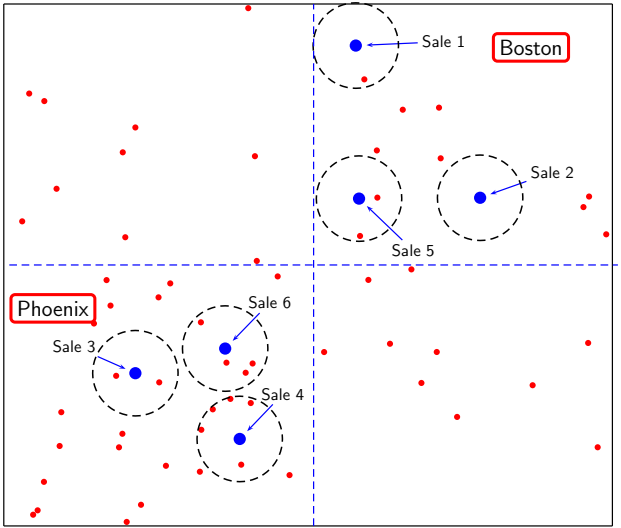

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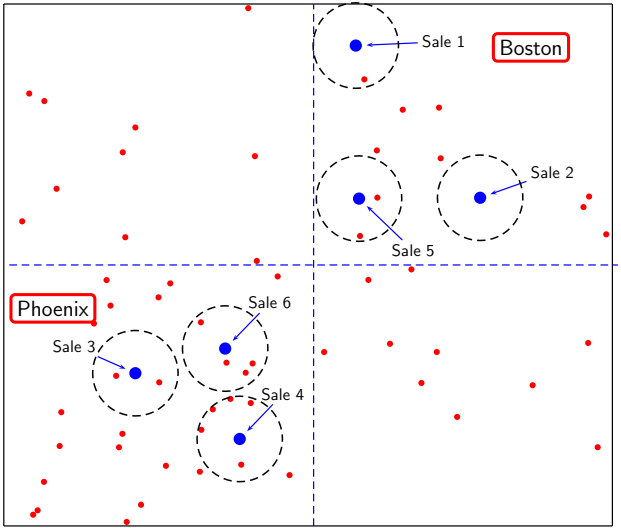

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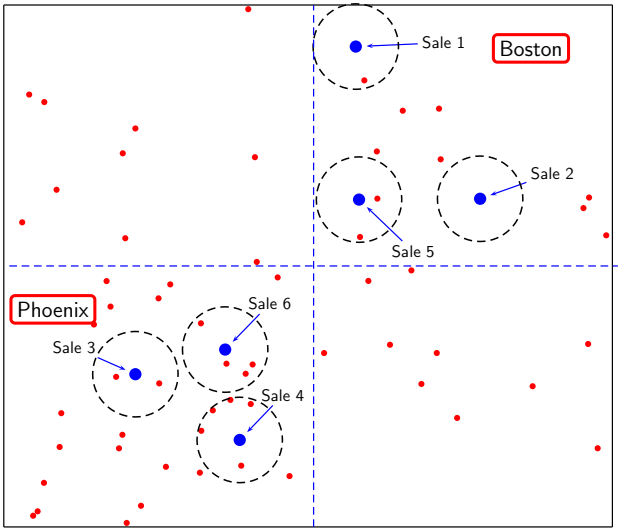

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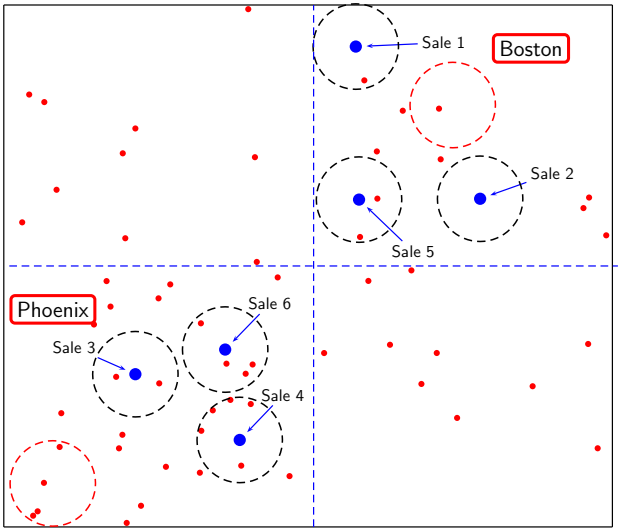

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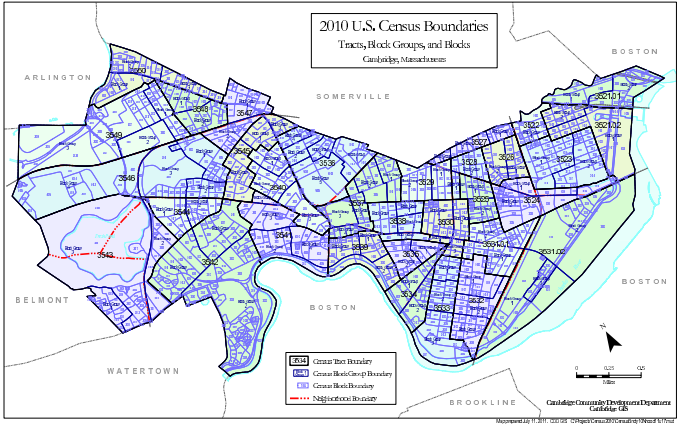
- And prices are lower.
- Draw any circle, on average you will get three times as many foreclosures in Phoenix.

Gerardi et al. (2015)

- (1) "Census Block Group Controls" in Cambridge, MA. (2010 Census)



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(3) Stocks versus flows

- Investment externalities:
  - Prior to foreclosure, borrower controls property but benefit of any investment accrues to lender.
  - Post foreclosure, lender controls property but gets only limited benefit from improvements.
- *Stock* of distressed property matters. Not flow.
- Public records show only flows. Can figure out stock of REO.
- Stock of distressed property more difficult.
  - We use proprietary data to identify properties with delinquent mortgages.

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  - Payment history of borrower.
  - 12-15% of all foreclosures.

Results

- Effect of REO/SDQs by Distance based on 2009 Sample:

	Difference in number of:			
	Long SDQs	REO inventory	REO sold in:	
			Last year	1-2 years ago
0-100 yards	-0.015	-0.014	-0.010	-0.004
	(15.8)	(16.2)	(19.8)	(4.6)
101-200 yards	-0.007	-0.007	-0.003	0.001
	(12.1)	(12.7)	(9.8)	(.9)
201-300 yards	-0.001	-0.002	-0.001	0.002
	(4.9)	(7.8)	(3.2)	(8.6)
301-600 yards	-0.002	-0.001	-0.001	0.001
	(21.0)	(8.1)	(8.6)	(11.75)

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  - Water view is *always* more valuable.

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- Condition results.
- Fact that the effect appears when borrower becomes DQ and goes away when new owner has time to invest.

The slide you've all been waiting for...

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- The end.

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